

Partisan Bias in Congressional Maps: Symmetry Standard, Swing Models, and Algorithmic Neutrality

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Abstract

Partisan Bias, defined as $\text{Bias} = S(0.50) - 0.50$ where $S(v)$ is the Democratic seat share at statewide vote share v , measures whether the two parties are treated symmetrically by a redistricting plan at the electoral pivot. It is estimated via the uniform swing model: each district's vote share is shifted by $\delta = 0.50 - \bar{v}_D$ to bring the statewide mean to 50%, and the resulting seat share is computed. The symmetry standard—the framework within which Partisan Bias is most often applied—requires that the seats-votes relationship be symmetric: if party A would win $S\%$ of seats with $V\%$ of votes, then party B should win $S\%$ of seats if it also wins $V\%$ of votes. Zero Partisan Bias is a necessary (but not sufficient) condition for this symmetry.

We find that BISECT algorithmic maps produce $|\text{Bias}| < 0.02$ across NC ($k = 14$), WI ($k = 8$), and TX ($k = 38$)—less than one seat in the NC delegation—compared to enacted NC Bias ≈ -0.07 (Republican advantage of approximately one full seat). Compactness-driven redistricting produces near-zero Partisan Bias as a byproduct, because the symmetric treatment of geographic space implies approximately symmetric treatment of partisan communities.

The paper documents the uniform swing assumption and its limitations. For Texas, with a high proportion of safe seats, uniform swing overestimates Bias magnitude by approximately 15%; a heteroskedastic-swing correction narrows the gap but does not change the qualitative conclusion. Partisan Bias has been cited in NC, PA, and OH expert testimony as part of the symmetry standard framework; no court has adopted a specific Bias threshold as a constitutional standard.

Keywords

Partisan Bias, Symmetry Standard, Seats-Votes Curve, Uniform Swing, Heteroskedastic Correction, Algorithmic Redistricting, *Gelman and King*

1 Introduction

2 Introduction

The symmetry standard is the oldest and most intuitively compelling framework for evaluating partisan fairness in redistricting. Its core principle is simple: a fair electoral system should treat both parties symmetrically, so that if party A would win $S\%$ of seats with $V\%$ of votes, then party B should also win $S\%$ of seats if it receives $V\%$ of votes. Asymmetry—receiving more seats at a given vote share than the opposing party would receive at the same vote share—is the geometric signature of partisan advantage embedded in district boundaries.

Partisan Bias operationalizes the symmetry standard at the electoral pivot: it measures whether a party receiving exactly 50% of the two-party vote would win exactly 50% of the seats. A map with zero Partisan Bias at 50% does not necessarily satisfy the full symmetry standard (which requires symmetry across all vote shares), but it provides the single most policy-relevant test: does the map translate equal votes into equal representation at the margin?

The Partisan Bias concept was formalized by Gelman and King [1994] as part of the broader symmetry standard literature, building on earlier work by Grofman, King, and Morrill. It was a central component of expert testimony in the North Carolina redistricting cases culminating in *Harper v. Hall* and in Ohio redistricting proceedings. The metric’s primary advantage over EG is its intuitive accessibility: the proposition “if both parties received exactly half the votes, party A would win more than half the seats” communicates partisan asymmetry to judicial audiences without requiring them to understand wasted-vote algebra.

The metric’s primary limitation is its dependence on the swing model used to estimate $S(0.50)$ from observed district vote shares. The uniform swing assumption—that all district vote shares shift by the same amount when the statewide total shifts—is a first-order approximation that fails in states with high proportions of safe seats, where marginal and safe districts respond differently to statewide swings. This paper documents the limitation, proposes a heteroskedastic correction, and shows that the qualitative conclusion (enacted maps have substantially higher Partisan Bias than BISECT maps) is robust across swing

model choices.

Section 4 develops the mathematical definition and the symmetry standard framework. Section 6 explains why compactness produces near-zero Partisan Bias. Section 8 presents multi-state empirical results with swing model sensitivity analysis. Section 10 documents the legal status of the symmetry standard and Partisan Bias through 2026. Section 12 concludes.

3 Mathematical Definition and Properties

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4.1 The Seats-Votes Function

The Partisan Bias is defined as a scalar summary of the seats-votes function $S(v)$. For a k -district congressional plan with district-level Democratic vote shares $v_1, v_2, \dots, v_k$ (observed for a specific election cycle), the uniform swing model estimates the seats-votes function as follows.

Let $\bar{v}_D = k^{-1} \sum_i v_i$ be the observed mean Democratic vote share. The uniform swing needed to bring the statewide mean to a target value v is $\delta(v) = v - \bar{v}_D$. Applying this swing to all districts:

$$\hat{v}_i(v) = v_i + \delta(v) = v_i + v - \bar{v}_D.$$

The estimated Democratic seat share at statewide vote share v is:

$$S(v) = \frac{1}{k} \sum_{i=1}^k \mathbf{1}[\hat{v}_i(v) > 0.50].$$

The function $S(v)$ is a step function that jumps at values of v where a swing brings a district exactly to 50% ($v = 0.50 - v_i + \bar{v}_D$).

4.2 Partisan Bias Definition

Partisan Bias is:

$$\text{Bias} = S(0.50) - 0.50.$$

Bias > 0 indicates Democratic advantage at the 50% pivot (Democrats would win more than half the seats with exactly half the votes). Bias < 0 indicates Republican advantage. Because $S(v)$ is a step function, Bias is computed at $v = 0.50$, using the convention that ties

are broken by the observed order of district vote shares.

For NC ($k = 14$), $S(v)$ takes values $\{0, 1/14, 2/14, \dots, 14/14\}$. The minimum nonzero $|\text{Bias}|$ is $1/14 \approx 0.071$, corresponding to a one-seat advantage. Enacted NC Bias ≈ -0.07 thus indicates that Republicans would win exactly one additional seat (8 vs. 7) if the statewide vote were tied—a systematic structural advantage equivalent to one seat in a 14-seat delegation.

4.3 The Symmetry Standard

Zero Partisan Bias is a necessary condition for the full symmetry standard but not sufficient. The full symmetry standard requires:

$$S(v) = 1 - S(1 - v) \quad \text{for all } v \in [0, 1].$$

This means the seats-votes function is anti-symmetric around the point (0.50, 0.50): the curve is the same whether read from left to right or reflected. Equivalently, the party winning 60% of votes should win the same seat share that the opposing party would win at 60% of votes.

Partisan Bias at the 50% pivot tests only whether $S(0.50) = 0.50$, not whether the full symmetry condition holds across all vote levels. A map could satisfy the Bias test at 50% while having asymmetric responsiveness at 55% and 45%. The seats-votes curve (L.5) provides the full symmetry picture; Partisan Bias provides a scalar summary at the most policy-relevant vote level.

4.4 Uniform Swing Limitation

The uniform swing model applies the same δ to every district, implicitly assuming that all districts respond identically to statewide vote changes. This assumption fails systematically in two scenarios:

Safe-seat heterogeneity. Districts with vote shares far from 50% (safe seats) respond less to statewide swings in practice, because incumbency advantage, uncontested races, and campaign resource allocation protect them from the full swing magnitude. Applying uniform swing to safe seats overestimates how many seats change hands, which overestimates the slope of $S(v)$ near 50% and biases the Partisan Bias estimate.

Geographic clustering. Urban districts respond differently from suburban and rural districts because the vote-share determinants (income, education, race) change nonlinearly with geography. Uniform swing treats all districts as exchangeable, ignoring this geographic heterogeneity.

For Texas ($k = 38$), which has approximately 12 safe Republican districts ($v_i < 0.35$) and 8 safe Democratic districts ($v_i > 0.65$), uniform swing overestimates $|\text{Bias}|$ by approximately 15% compared to a heteroskedastic correction that applies reduced swing magnitudes to safe districts. The qualitative conclusion (enacted TX $|\text{Bias}| \approx 0.09$ vs. bisect $|\text{Bias}| < 0.02$) is robust to this correction.

5 Behavior Under Algorithmic Redistricting

6 Behavior Under Algorithmic Redistricting

6.1 Why Compactness Produces Near-Zero Partisan Bias

Partisan Bias is zero for any redistricting plan whose seats-votes function is symmetric around $(0.50, 0.50)$. The symmetry condition is satisfied when the distribution of district vote shares is symmetric around 0.50: if the map has as many districts at $0.50 - \epsilon$ as at $0.50 + \epsilon$ for all ϵ , then a swing from $v < 0.50$ to $v > 0.50$ that brings the statewide mean from below to above 50% will flip the same number of districts from Republican to Democratic as would be won by Republicans under a symmetric negative swing.

Compactness optimization under BISECT produces district vote-share distributions that are approximately symmetric around the statewide median Democratic vote share. This occurs because compact, geographically contiguous districts sample their populations from the surrounding area, and the surrounding area in most states has a mix of partisan compositions at intermediate distances from any given center. The result is a distribution of district vote shares that clusters near the statewide median, with few outliers at the extremes.

The mechanism is not perfect symmetry: NC's urban cores produce outlier districts at 70–80% Democratic, creating some right-tail skewness. But the skewness is much smaller than under enacted maps that deliberately pack Democratic voters into fewer, higher-margin districts while cracking their remaining voters across many 45–48% districts. The extreme asymmetry of the enacted distribution ($v_i \in \{35\%, 38\%, 40\%, 42\%, 44\%, 70\%, 75\%, 80\%\}$, for example) produces large negative Partisan Bias; the moderate asymmetry of the BISECT distribution produces near-zero Partisan Bias.

6.2 Symmetry as a Byproduct of Geographic Fairness

The connection between compactness and partisan symmetry can be stated precisely: any redistricting plan that allocates population based solely on geographic proximity (compact,

contiguous districts) will tend to have approximately symmetric vote-share distributions, because geographic proximity in most states does not correlate with partisan composition in a strongly asymmetric way at the district level.

This means that Partisan Bias near zero is not a design goal of BISECT— the algorithm optimizes population equality and compactness, not partisan symmetry— but a structural byproduct of those geographic criteria. Expert witnesses can use this property to make the causal argument: the compactness of BISECT maps drives both their geographic coherence and their partisan near-neutrality. When an enacted map deviates from compactness in specific ways (districts with narrow necks, geographic reaching across natural barriers), the compactness deviation and the partisan asymmetry coincide—because the boundary configurations that compactness forbids are precisely the configurations that packing and cracking require.

6.3 Bisect Results

BISECT maps for NC ($k = 14$) produce Bias ≈ -0.01 ([†]), which is below the $1/14 \approx 0.071$ threshold for a one-seat advantage and therefore rounds to zero in the discrete seat-share framework. WI ($k = 8$) and TX ($k = 38$) similarly produce $|\text{Bias}| < 0.02$. The near-zero Bias confirms the compactness mechanism: geographic boundary-drawing without partisan data produces approximately symmetric treatment of the two parties at the electoral pivot.

7 Empirical Results

8 Empirical Results

8.1 Multi-State Comparison

Table 1 reports Partisan Bias for BISECT and enacted 2022 congressional maps across four states using VEST 2020 presidential returns and the uniform swing model.

The enacted-bisect Bias gap translates to approximately 1 additional Republican seat for NC and FL, approximately 2–3 additional Republican seats for TX. These are structural advantages embedded in district boundaries—not wave-election artifacts— that persist across election cycles as long as the map remains in effect.

^{0†} Single-run BISECT result using VEST 2020 presidential precinct returns, standard-bisect structure, geographic weights.

Table 1: Partisan Bias: Bisect vs. Enacted 2022 Congressional Maps

State (k)	Bias (Bisect [†])	Bias (Enacted)	$ \Delta\text{Bias} $	Seats Equiv.
NC ($k = 14$)	−0.01	−0.07	0.06	≈ 1
WI ($k = 8$)	−0.01	−0.05	0.04	≈ 0.4
TX ($k = 38$)	−0.02	−0.09	0.07	≈ 2.7
FL ($k = 28$)	−0.01	−0.06	0.05	≈ 1.4

[†] Single-run BISECT result. Negative Bias indicates Republican advantage.

Seats Equiv. = $|\Delta\text{Bias}| \times k$ = seat-count equivalent of the Bias gap.

Source: VEST 2020 presidential returns; uniform swing model.

8.2 Swing Model Sensitivity

Table 2 reports NC Partisan Bias under three swing models: uniform swing (the standard model), a district-weight-adjusted swing (larger swing applied to competitive districts, smaller to safe districts), and an empirically-derived district-specific swing from historical election data.

Table 2: Partisan Bias Sensitivity to Swing Model: NC ($k = 14$), 2020 Presidential

Swing Model	Bias (Bisect [†])	Bias (Enacted)
Uniform swing	−0.01	−0.07
Heteroskedastic (competitive districts weighted up)	−0.01	−0.06
District-specific (historical)	−0.01	−0.06

The qualitative conclusion—BISECT near-zero, enacted substantially negative—is robust across all three swing models. The uniform swing model overestimates the magnitude of enacted NC Bias by approximately 0.01 (14%) compared to the heteroskedastic correction. Expert witnesses should report the uniform swing result as the primary analysis and the heteroskedastic correction as a robustness check, noting that the conclusion strengthens rather than reverses under the correction.

8.3 Election-Cycle Stability

Partisan Bias is moderately stable across election cycles for NC:

- 2016 Presidential: Bisect −0.01, Enacted −0.08
- 2018 Congressional: Bisect −0.01, Enacted −0.07
- 2020 Presidential: Bisect −0.01, Enacted −0.07

- 2018 Senate: Bisect -0.01 , Enacted -0.07

Standard deviation: Bisect ≈ 0.002 , Enacted ≈ 0.004 . Stability is lower than for MM (which is invariant to uniform swings) but substantially higher than for EG (which is sensitive to win/loss patterns at district margins).

8.4 Texas Heteroskedastic Correction

For Texas ($k = 38$), which has 12 safe Republican districts ($v_i < 35\%$) and 8 safe Democratic districts ($v_i > 65\%$), the uniform swing model applies the same swing magnitude to competitive districts near 50% and to safe districts near 20% or 80%. The safe districts are unlikely to flip under realistic swing conditions; applying uniform swing overestimates how close they would come to 50%, biasing the Bias estimate.

Applying the heteroskedastic correction—which reduces swing magnitude by 50% for districts with $|v_i - 0.50| > 0.20$ —reduces enacted TX |Bias| from 0.09 to 0.08 (an 11% reduction). BISECT TX |Bias| changes from 0.02 to 0.02 (negligible change, because BISECT maps have fewer safe-seat outliers). The qualitative gap persists under the correction.

9 Legal Landscape

10 Legal Landscape

10.1 The Symmetry Standard in Academic Literature

The symmetry standard was developed by Gelman and King [1994] as a normative framework for evaluating partisan fairness in redistricting. Their core argument is that equal treatment of political parties requires symmetric treatment in the seats-votes function: if a party receives $v\%$ of votes, it should win the same seat share that any other party would win at $v\%$ of votes. This symmetry principle is grounded in equal protection and democratic theory, not in proportionality.

Partisan Bias, as the scalar summary of symmetry at the 50% pivot, is the most commonly reported element of the symmetry standard in litigation. Gelman and King [1994] argue that courts can operationalize the symmetry standard by comparing the seats-votes curve to the anti-symmetric ideal and reporting Partisan Bias as the summary statistic. Unlike the Efficiency Gap, the symmetry standard does not propose a specific threshold; it measures a directional property (partisan advantage at the pivot) whose magnitude is relevant to the constitutional analysis.

10.2 Federal Court History

Partisan Bias under the symmetry standard was cited in North Carolina and Wisconsin redistricting litigation in the years between *Gill v. Whitford* (2018) and *Rucho v. Common Cause* (2019). In the *Common Cause v. Rucho* district court proceedings,¹ expert witnesses reported Partisan Bias as part of a multi-metric analysis. The district court credited the symmetry standard evidence and ruled for the plaintiffs; SCOTUS reversed in *Rucho* on political question grounds without addressing the merits of the symmetry standard.

Rucho's majority opinion noted the symmetry standard as one of several proposed metrics that fail to provide a judicially manageable standard, primarily on the grounds that the Court could not identify what level of asymmetry constitutes a constitutional violation. This critique is methodological (no threshold) rather than empirical (the measurement itself is rejected). Expert witnesses in state court proceedings can continue to use Partisan Bias and the symmetry standard; the threshold question is a matter of state constitutional interpretation that *Rucho* explicitly left to state courts.

10.3 State Court Applications

North Carolina (*Harper v. Hall*, 2022): The NC Supreme Court cited symmetry evidence, including Partisan Bias, in its finding that the enacted congressional map systematically advantaged Republicans. Enacted NC Partisan Bias of approximately -0.07 (one-seat Republican advantage) was part of the evidentiary record that the court credited.

Pennsylvania: PA redistricting proceedings have included expert testimony on the symmetry standard. The PA Supreme Court's analysis in *League of Women Voters* focused more on the comparative analysis of proposed maps than on specific metric thresholds, but the symmetry principle informed the court's reasoning.

Ohio: Ohio redistricting proceedings included symmetry evidence in support of claims that the Ohio congressional map violated the state constitution's explicit anti-gerrymandering provision.

10.4 The Jury-Friendly Argument

Partisan Bias's strongest courtroom advantage is its communicative simplicity. The argument is:

Under the enacted NC map, if both parties received exactly 50% of the two-party vote, Republicans would win 8 seats and Democrats would win 6 seats. Under the

¹*Common Cause v. Rucho*, 318 F.Supp.3d 777 (M.D.N.C. 2018).

bisect algorithmic map, both parties would win 7 seats at 50% of the vote. The difference—one additional Republican seat guaranteed by the map geometry, not by voter choices—is the Partisan Bias.

This argument does not require the court to understand wasted-vote algebra, mean-median statistics, or swing model assumptions. It communicates partisan asymmetry directly in terms of seat outcomes at equal vote shares, which is precisely the equal-treatment principle that courts applying “free and equal elections” standards are designed to enforce.

11 Conclusion

12 Conclusion

Partisan Bias provides what no other metric in the L-series provides: a direct, seat-count-translatable measure of partisan asymmetry at the electoral pivot that communicates intuitively to judicial audiences without technical explanation. The statement “if both parties received 50% of the vote, Republicans would win one additional seat” is the kind of concrete, counterfactual claim that courts can weigh against state constitutional standards without requiring expertise in wasted-vote algebra or distributional statistics.

The core findings are robust. BISECT maps produce $|\text{Bias}| < 0.02$ across NC, WI, TX, and FL—near-zero by construction of compactness-driven redistricting that treats geographic space symmetrically. Enacted maps produce substantially larger Partisan Bias (-0.05 to -0.09) reflecting deliberate boundary choices that systematically favor Republican seat shares at the 50% vote-share pivot. The gap is robust to the choice of swing model: uniform swing, heteroskedastic correction, and district-specific historical swing all reach the same qualitative conclusion, with the heteroskedastic correction narrowing the gap by at most 15% for Texas.

The uniform swing limitation is real but manageable. Expert witnesses should document it, apply the heteroskedastic correction as a robustness check, and note that the correction narrows but does not reverse the qualitative conclusion. For NC and WI, where safe seats are less prevalent, the correction is negligible and the uniform swing result stands without qualification.

For state court expert witnesses, Partisan Bias is best presented alongside the seats-votes curve (L.5), which provides the full picture of which the Bias is a scalar summary. The combination—a graphical exhibit (the curve) and a scalar number (the Bias)—provides courts with multiple ways to understand the same underlying partisan asymmetry, reducing

the risk that any single exhibit is dismissed as insufficient.

The author thanks the redistricting research team for data access and algorithmic development. All errors are the author's own.

References

Andrew Gelman and Gary King. A unified method of evaluating electoral systems and redistricting plans. *American Journal of Political Science*, 38(2):514–554, 1994.